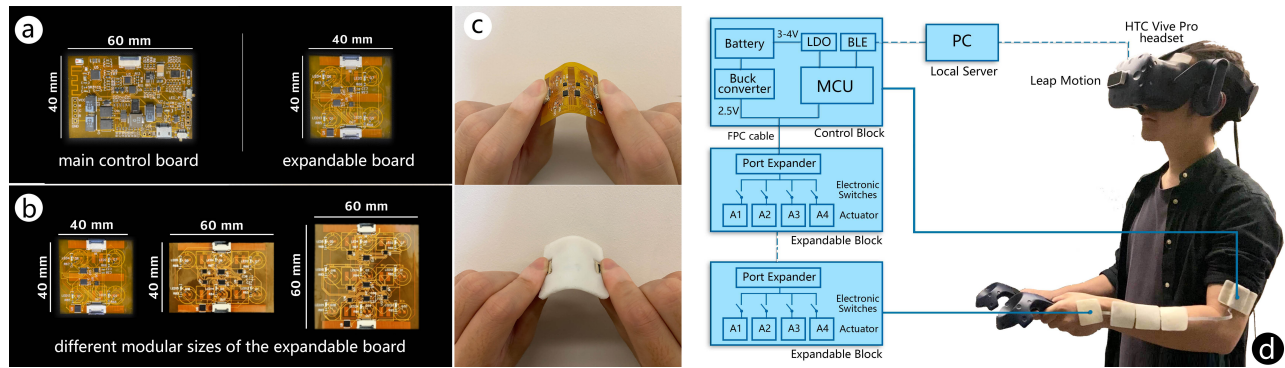






**Figure 2: (a) Main control board (left) and multiple expandable boards (right) of the device. (b) Three different modular sizes of the expandable board. (c) The electronic devices have properties similar to physical skin. (d) System diagram of SoftVibe.**

control circuit wirelessly communicates with the VR workstation, allowing control of up to 26 actuators. The software interface enables users to create vibrotactile feedback by adjusting parameters and providing predefined templates. A multi-timeline feature facilitates synchronized visual, auditory, and vibrotactile feedback. *SoftVibe* enhances user immersion and expands haptic interfaces across the body, offering designers a comprehensive and user-friendly toolkit for exploring multisensory VR design possibilities (Fig. 1b).

## 2 SOFTVIBE TOOLKIT

*SoftVibe* is a toolkit to support the rapid-prototyping process of on-body vibrotactile feedback in VR using the silicon-based flexible electronics. Fig. 1 shows the diagram of *SoftVibe* which contains both software and hardware components.

### 2.1 Hardware

This system utilizes the skin-integrated electronics [Jung et al. 2021; Yu et al. 2019] to transmit sensory stimuli wirelessly using a vibrating actuator that converts energy into vibrations (Fig. 2d). Skin-integrated electronics combine various electronic components (e.g., electronic circuits, chips, electrodes, etc.) on a flexible substrate to mimic human skin, resulting in devices with characteristics similar to physical skin, such as stretchability, bendability, and deformability [Jung et al. 2021] (Fig. 2c). Unlike traditional rigid electronic devices, flexible electronic skin offers enhanced flexibility to meet users' diverse deformation requirements.

### 2.2 Software

The *SoftVibe* toolkit's software interface is an immersive VR application developed with Unity3D 2019. It enables loading 3D models and creating visual effects such as trajectory animation and particle systems in VR. The graphical user interface (GUI) allows designers to author vibrotactile signals in different patterns to control flexible on-skin vibrotactile modules. Users can feel real-time vibrotactile feedback for all signal settings in the software.

## 3 CONCLUSION

In this paper, we present *SoftVibe*, a toolkit to support the usage of flexible electronics for creating vibrotactile feedback in VR. With

*SoftVibe*, we aim to facilitate and popularize the usage of flexible electronics in immersive VR.

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